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PALEOMAGNETIC EVIDENCE FOR A CRETACEOUS DISTURBING EVENT IN THE PRECAMBRIAN JOHNNY LYON GRANODIORITE, COCHISE COUNTY, ARIZONA[1]

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ABSTRACT

Block samples were collected in an east-west profile across the 1,655 m.y.-old Johnny Lyon granodiorite, Cochise County, Arizona. Progressive alternating field demagnetization shows that the NRM directions are stable to 175 Oe, and remain tightly grouped. Beyond this field the angular changes are large and irregular. The mean directions initially and after 175 Oe are remote from either the axial or the inclined present fields. The corresponding paleomagnetic pole after 175 Oe shows good agreement with poles from Cretaceous rocks from North America. This independent evidence supports the conclusion from isotopic studies (Silver and Deutsch 1963) that the granodiorite was profoundly affected by Cretaceous alteration processes.

INTRODUCTION

One of the objects in studying the natural remnant magnetization (NRM) of rocks is to determine dated paleomagnetic pole positions. For Precambrian rocks, strict criteria must be applied to ensure that the magnetization is both stable and reliable. Moreover, the geologic setting is of fundamental importance, for if the rocks have been disturbed structurally or subjected to metamorphism or alteration since their formation, their magnetization may have been permanently altered. Data from isotopic studies can be used to detect an alteration history in the rock. This note concerns a paleomagnetic study of a Precambrian intrusion which has been shown from isotopic evidence to have been profoundly affected by late Mesozoic alteration processes.

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GEOLOGIC SETTING

Rocks in the Dragoon Quadrangle, Cochise County, Arizona range in age from Precambrian to Recent. Detailed studies have been reported by Cooper and Silver (1964). They describe a large, postkinematic, granodiorite pluton in the Johnny Lyon Hills, which is in sharp intrusive contact with the older Precambrian Pinal Schists. All the Precambrian rocks in this area are truncated unconformably by sediments of the younger Precambrian Apache Group. Along the eastern slopes of the Johnny Lyon Hills these sediments form an arcuate pattern around the granodiorite and dip between 40° and 65° to the northeast. Cooper and Silver (1964) comment that if the pre-Apache Group erosion surface is rotated to the horizontal, then the granodiorite–Pinal Schist contacts more reason-

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ably assume the attitude of roof pendants.

Silver and Deutsch (1963) summarize their very detailed Uranium/lead isotopic studies on zircons from the granodiorite which indicate an age of crystallization of 1,655 ± 20 m.y. for the intrusive. The data further suggest that extensive alteration processes affected the mineral systems in late Mesozoic–early Cenozoic time at 90 m.y.

SAMPLE COLLECTION AND PREPARATION

Five block samples, each weighing about 15 lb, were collected at half-mile intervals along the Willcox to Cascabel road, west of the Cross X Ranch (110.1° W, 32.1° N). The profile lies close to one of the sites (L312) chosen by Silver and Deutsch (1963) for their isotopic study so as to be as free as possible from detectable alteration effects. A tripod plate device was used to orient the blocks, and azimuth was measured with a magnetic compass: a declination of 13° E was read from the Isogonic Chart for the United States, 1965.0 (U.S. Coast and Geodetic Survey). In the laboratory, cylinders of 1-inch dimension were drilled from each block: a total of 17 specimens was used in the study.

DEMAGNETIZATION STUDIES

The direction and moment of the NRM of each specimen were measured on a PAR Spinner Magnetometer, Model SM1 at the University of Texas at Dallas. Alternating field (AF) demagnetization experiments were conducted on equipment described by Helsley (1969). In a shielded μ-metal room, the specimens were tumbled about three axes inside a coil in which smoothly decreasing fields with peak intensities of 88, 175, 350, 700, and 1,400 Oe were produced. This procedure was applied to one specimen from each block sample.

The demagnetization paths for three of the specimens are shown in figure 1. The response for all five specimens as far as the 175-Oe stage is consistent: there is very little change in the NRM directions, and they remain fairly tightly clustered. Beyond this field, the angular changes are large and irregular and show no tendency to converge.

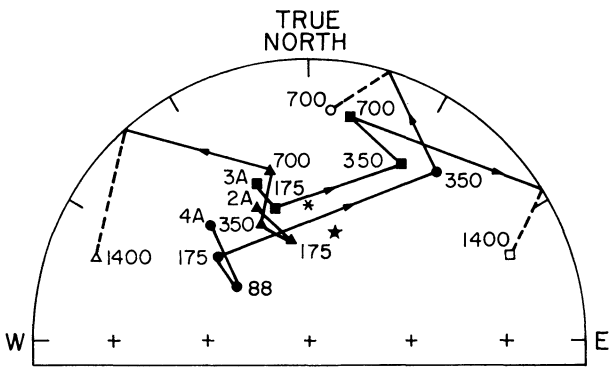


FIG. 1.—Demagnetization paths for three specimens plotted on an equal area projection. Solid symbols are north seeking poles downward; open symbols are north seeking poles upward. Asterisk is the average axial field; star is the field observed today. Alternating field stages are in oersteds.

The important observation from these preliminary demagnetization experiments is that at no stage do the NRM directions group around either the axial or the inclined present earth's magnetic field. Thus, although the NRM is formed mainly of components of low coercive force, these are capable of retaining a stable magnetization which does not apparently reflect the present field axis.

SUPPORTING STUDIES

Opaque mineralogy studies show that the main iron-titanium oxide is titanomag-

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netite (grain size up to 30 μ) which is directly oxidized to hematite in amounts up to 75 percent. All the titanomagnetites are oxidized to hematite marginally and as broad wedges along 111 planes; some are almost completely replaced.

Many small fibrous, oxidized titanomagnetites occur in the mafic minerals, commonly being parallel to the cleavage directions of biotite. They may indicate a second generation of spinel. In support of this there is a good deal of sphene around the margins of all the spinels, and hematite reddening occurs throughout the rock.

A Curie point determination in a field of 1,500 Oe indicates a spinel phase as the only carrier of the saturation remanence, which is interesting in view of the abundant hematite observed microscopically. The spinel has a Curie point of 580°–595° C, and the cooling curve is reversible.

The intensity range of the 17 specimens in the collection is small, with a median at 4.6×10^{-5} emu/cubic centimeter. Their susceptibilities also show a small range, with a median at 13.0×10^{-5} emu/oersted per cubic centimeter. The ratio of the intensity to the susceptibility (the modified Koenigsberger ratio, Q′) has a median at 0.41. Ideally this ratio should be close to unity for stable rocks: however it is a factor of four above the value quoted by Irving (1964, p. 92) as indicating the presence of large unstable components of magnetization in a rock.

An artificial thermoremanence (TRM) was given to one specimen by cooling it from 750° C in a field of 0.3 Oe. The direction of TRM was within 2° of the applied field axis, which suggests that the rock is magnetically isotropic, and thus capable of recording the direction of a field in which it is cooled.

The conclusions from these supporting studies are that the specimens have not been affected by lightning (there are no anomalously high intensity values), that the magnetic content is uniform from sample to sample (the range in intensity and susceptibility is small), that a titanium free magnetite carries the saturation remanence (there is a sharp Curie point at 580°–595° C, and direct oxidation of the spinel to hematite), that the Q′ ratios suggest the NRM is not composed of large unstable components, and that the rock is capable of acquiring a magnetization which accurately reflects the applied field.

TABLE 1

MEAN DIRECTIONS OF MAGNETIZATION AND PALEOMAGNETIC POLES

Stage	Dec.	Inc.	N	R	k	a_{95}	Long.	Lat.	δp	δm
Initial...	341	39	17	16.704	54	5	133° E	70° N	3	6
175 Oe...	335	54	17	16.466	30	7	172° E	69° N	6	9

NOTE.—Dec. = declination of the north seeking magnetic vector, east of true north; Inc. = its inclination, positive downward; R = the length of N unit vectors, one per specimen; k = (N-1)/(N-R); a_{95} = the semiangle of the cone of 95 percent level confidence; Long. and Lat. = the longitude and latitude, respectively, of the paleomagnetic pole; δp and δm = the semiminor/major axes of the oval of 95 percent level confidence.

BLANKET DEMAGNETIZATION

To test further the hypothesis that a stable magnetization is present in the granodiorite, the remaining specimens were demagnetized to 175 Oe, because above this field the NRM directions no longer were well grouped. Mean directions initially and at 175 Oe were computed using Fisher's methods (1953). They a·e listed in table 1 and shown in figure 2 with the circles of 95 percent level confidence (a_{95}).

There is low dispersion of the initial NRM directions, and the directions after 175 Oe AF treatment are slightly more scattered. Nevertheless, the earlier suggestion, that the direction of the stable magnetization in the granodiorite is *not* related to the present field, is well shown in figure 2. Only four

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(out of 34) directions are within 10° of the axial or inclined present field axes.

The important question now arises: Is the divergence from the ambient field due to declination errors introduced by magnetic orientation of the samples in the field? This could only be answered by another collection using sun compass orientation

from several Cretaceous rocks from North America (taken from Larochelle 1968, table 4).

A comparison of the granodiorite pole at 175 Oe with poles from other Precambrian rocks from North America shows that it is quite remote from them (see summary by Spall 1970). However, the agreement be-

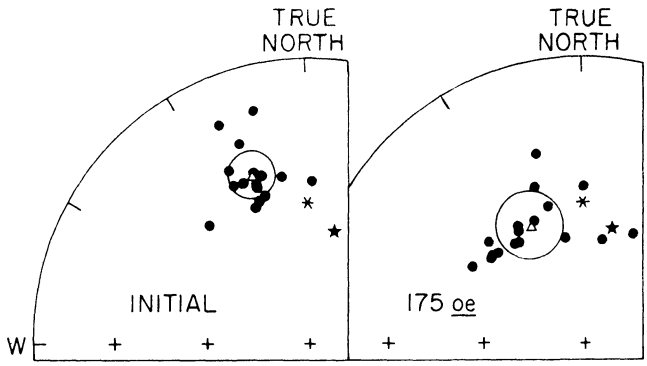


FIG. 2.—Directions of NRM initially and at 175 Oe, plotted on an equal area projection: all north poles are downward seeking. Triangles are mean directions with their associated circles of 95 percent level confidence (α_{95}). Asterisk is the axial field; star is the inclined field.

methods. However, two pieces of evidence suggest that declination errors are not responsible for the divergence from the present field. First, it would require a constant error of between 20° and 25° in measuring the local declination, giving a very high total declination of about 35° E over the 3-mile sampling profile. (The low intensity of the specimens implies that such an anomaly would not be due to the granodiorite.) Second, given such a large declination error, the initial NRM inclinations are then too shallow to permit them to enclose the axial or inclined field axes; further, the circle of confidence after 175 Oe would include only the axial field, so, given the declination error, the stable magnetization would have to be of Recent origin.

PALEOMAGNETIC POLES

Paleomagnetic pole positions offer a means of interpreting the mean magnetic directions obtained from the granodiorite. They are plotted in figure 3 together with the poles

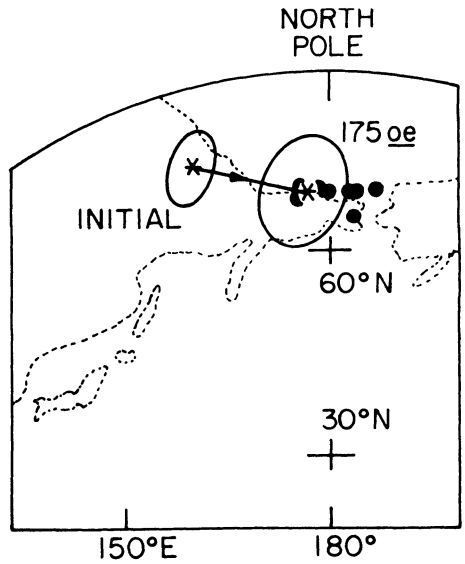


FIG. 3.—Mean paleomagnetic poles (asterisks) for the granodiorite, before and after AF demagnetization, with the associated ovals of 95 percent level confidence. Solid circles are poles for some Cretaceous rocks from North America (see text). Equal area projection.

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tween this granodiorite pole and the Cretaceous poles of figure 3 is rather striking. It implies that the memory of the Precambrian field direction has been obliterated. This independent evidence from paleomagnetism supports the conclusion of Silver and Deutsch (1963) that the granodiorite was profoundly affected by Cretaceous alteration processes.

No structural correction was applied in the above analysis. A correction applied for a pre-Apache Group erosion surface yields a pole (after the 175 Oe stage) at long 47° W, lat 65° N, which is considerably removed from poles from other Precambrian rocks of similar age (Spall 1970). Thus the explanation of a Cretaceous remagnetization of the granodiorite seems to be most plausible. However, the alternative explanation of a declination error and a relatively stable magnetization of Recent origin cannot be entirely dismissed with the evidence available.

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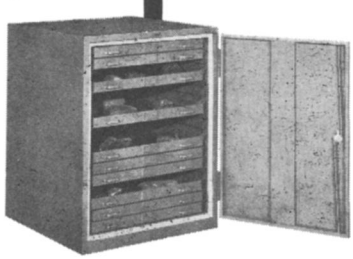
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